# The Lock: the $a_0$ Law as a Measurement, Form B and Its Rules, and the Budget the Cold Component Cannot Meet

Working note (PAPER73), version 1.0, 2026-11-11. AI-assisted research programme; not peer reviewed; not reproduced by any outside group. Companions: PAPER69 (DOI 10.5281/zenodo.24025170, the closure record), PAPER70 (DOI 10.5281/zenodo.24092073, field equations), PAPER71 (DOI 10.5281/zenodo.24095262, the switch class) and PAPER72 (v1.1 DOI 10.5281/zenodo.24125791, the form the record closes on). $\kappa = \frac{1}{2}$ is fitted. The theory is not closed. The cold component is a gravitating field component, never a particle. Nothing here says the data favour this framework.

**Abstract**

The non-baryonic mass is real, conserved and gravitating; there is no particle in it. What the data measure about it is one relation, and that relation is a measurement, not a hypothesis: around galaxies the enclosed non-baryonic-to-baryonic mass ratio follows $X(x) = \frac{1}{2}(\sqrt{1+4x^2} - 1)$ with $x = r/r_M$, to a worst bin median of 0.057 dex canonical and 0.021 dex alt over $x = 0.18$ to 10, against a pre-registered line of 0.10 (SHP1, 2700 points, 149 galaxies). Form B is the claim that this component is the cold, clustering sector of cosmology — cold dark energy — counted once, present in galaxies as the law's halo and present in hot systems at the cosmic share. This note states form B as six rules and then reports the result that ends it: **the amount cannot balance.** Two exact results cap what a halo can hold. T4: inside any ball whose boundary free-falls, the net polarization charge is exactly zero, at all times, for any momentum equation, so no charge is manufactured during infall. LFS1: for a standing clock the polarization is frozen, its charge moving only by order $|\Phi|/c^2 \sim 10^{-6}$, so a static halo cannot grow. The halo is therefore whatever charge the clock carried in, and the record's own infall dynamics deliver $1.06\,M_b$ while the law's halo needs $41.5\,M_b$ at the inner edge of KiDS's window and $97.8\,M_b$ at its middle edge, against a cosmic budget of $5.41\,M_b$ that is already spent at 38 kpc. KiDS rejects the cosmic share on its own, at $+4.85/+4.93\sigma$. The last flank — that the missing amount is really missing baryons in $g_{\rm bar}$ — is closed today by GAS1: the gas the reading needs is a fixed column $\Sigma_{\rm gas} = a_0/(\pi G x_b^2) = 6.16/7.45\,M_\odot\,{\rm pc}^{-2}$ out to 2.6 Mpc, and putting it into $g_{\rm bar}$ moves the outer amplitude to $0.039 \pm 0.006/0.032 \pm 0.006$ against the record's 0.891 gate. The reading destroys the fit it was invoked to save. What survives is stated last, with what to do about it.

## 1 What the framework says exists

This is the ontology, stated as the claim and not as a possibility.

1. Gravity carries one extra scale, $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$, equivalently $\Lambda = 32\pi a_0^2/c^4$.
2. The universe carries a cold, clustering component of 5.41 units of gravitating mass per unit of baryons ($\Omega_c/\Omega_b$, fitted to the acoustic peaks). It is not matter made of particles; it is a field component with $w \approx 0$.
3. Matter couples to the metric only. There is no fifth force on matter and no external field effect on real mass.
4. Where the component sits around baryons it obeys the law, and the law is a relation the data return (Section 2), not an equation of motion for a substance.
5. Its amount in a system is what that system's region brought, and amounts are conserved (Section 3, rule 5).

Two footings are carried throughout and never pooled: canonical $a_0 = 9.3603 \times 10^{-11}$ and alt $1.1312 \times 10^{-10}\,{\rm m\,s^{-2}}$; pairs of numbers are canonical/alt unless a row says otherwise.

## 2 The law as a measurement

The statistic is SHP1's. $X = M_C(<r)/M_b(<r) = g_{\rm obs}/g_{\rm bar} - 1$, the enclosed non-baryonic over the enclosed baryons, against $x = r/r_M$ with $r_M = \sqrt{GM_b/a_0}$; it is scored as $Y = \log_{10}(1+X)$, so an offset is dex in $g_{\rm obs}$ at fixed $g_{\rm bar}$. The law's prediction is $X(x) = \frac{1}{2}(\sqrt{1+4x^2} - 1)$: $\simeq x^2$ inside $r_M$ (density $\sim 1/r$), $\simeq x$ outside it (density $\sim 1/r^2$). This is the counted-once law, certified in Lean.

**SPARC.** SPARC has 2700 points in 149 galaxies. Its reduction is reproduced exactly (175 files, 2700 points against Lelli+2017's 2693, and 149 galaxies against 153). The law's shape is followed in every bin. The worst bin median is within 0.057 dex canonical (0.021 alt) for $x$ from 0.18 to 10, against a line of 0.10. Independently, form B passes SPARC rotation curves at rms 0.0563 (line 0.0660) canonical and 0.0558 alt, with no galaxy truncated inside its last measured point (median truncation radius 186 kpc).

**KiDS.** On B21's isolated lenses the data have the law's shape and sit above it: $1.40 \pm 0.03/1.28 \pm 0.03$ times the law, roughly flat from $x \sim 15$ to $\sim 200$ ($p$ of order $10^{-33}/10^{-21}$). On the host-free subset with every catalogued neighbour carrying the same share, form B **passes** its frozen gate at the $1.6\,R_h$ veto, for the share $X = 59.49/59.50\,M_b$; the measured outer amplitude there is $1.4002 \pm 0.2211/1.2740 \pm 0.2012$ of the law.

That is the certified accuracy: 0.057/0.021 dex on 2700 rotation-curve points, and a pass on KiDS host-free lenses whose error bar is $\pm 0.22/\pm 0.20$ of the law's level. Both are quoted, not rounded, and both are reproduced by PAPER73_audit.py from the lane files named beside them.

# 3 Form B, in six rules

1. **Fields.** The metric, one clock field that picks a slicing, and a polarization on its slices. Matter couples to the metric only. The action is QX17's (A), certified (label CERTIFIED).
2. **One conserved charge.** The clock equation is a conserved current. The universe carries 5.41 units of its charge per unit of baryons, a fitted number.
3. **Falling clock.** Where the clock free-falls with the expansion the charge is cold dust. This is the cold component of cosmology.
4. **Standing clock.** Where the clock stands still around baryons its own equations give the law, with $\kappa = \frac{1}{2}$ fitted. The halo is the charge itself and gravitates and lenses as mass.
5. **Counted once.** A halo is made of the charge its region brought. A hot system holds 5.41 times the baryons that fell into it. A body that brought no charge has no halo.
6. **The joint.** How a falling clock becomes a standing one is not derived.

Rules 1 and 2 are certified or fitted; rules 3 to 5 are two solutions of one action; rule 6 is the missing equation, and the hot branch of rule 5 is a rule and not a solution of the action.

# 4 The lock: the amount cannot balance

## 4.1 T4, the conservation cap (exact)

Inside any ball whose boundary free-falls, the net polarization charge is exactly zero, at all times, for any exchange rate, alignment and multipole — and this holds *for any momentum equation*, variational or not, provided the Legendre map is kept. It is Gauss's theorem, not dynamics, so no candidate is exempt from it by changing the dynamics. Its two landed records are: **T4 GAUSS (exact; SMF1 S0b reproduced)** (`toe_2026_11_10/AKS1/FROZEN_CRITERIA_T.md`) and **S0b (exact, Gauss)** (`closure_2026_11_10/SMF1/RESULT.md`), where the identity is the statement that in a region bounded where the frame free-falls the frame's polarization charge is zero at every time, so in net the exchange feeds only the frame's dust part and the boundary fluxes. The consequence is one line: **a halo's polarization charge cannot be manufactured during infall.**

## 4.2 LFS1, the standing halo is frozen (exact)

LFS1 asks whether the clock's leafwise flux can supply the halo and returns **ILL-POSED**. The flux is not a separate supply; it is the normal equation's own bookkeeping, and it adds no charge. Spherically, the polarization flux through a sphere is carried with the clock's flow, so with a standing clock it is frozen, and the enclosed charge inside a sphere that moves with the clock changes only by a fraction of order $|\Phi|/c^2$, about $10^{-6}$. The lane states the consequence plainly: **a static halo cannot grow.** (Its other two failures, the edge and the match, do not depend on its amendment: the static region ends where the clock falls freely, so its total polarization charge is zero, and the match to the expanding clock outside is a wall.)

T4 and LFS1 together are the cap, and it is not a claim about a mechanism:

*the halo's charge equals the dust charge the clock carried in.*

## 4.3 AMT1, the budget

Three amounts contend, and the arithmetic is printed in `closure_2026_11_10/AMT1_amount_budget/amt1_budget.out` (15 recorded cross-checks reproduce, worst relative miss $3.1 \times 10^{-3}$).

| Amount | Value (as recorded) |
|---|---|
| DEMAND | $M_c(< r) = M_b\, X(x)$; the share that passes KiDS is $X_s = 59.5\, M_b$ (PD1, ONE1) |
| BUDGET | $\mathbf{5.41\, M_b}$ ($\Omega_c/\Omega_b$, the whole cosmic cold-to-baryon ratio) |
| DELIVERED | $\mathbf{1.06\, M_b}$ turned around by $z = 0$; $1.4\, M_b$ settled (VSH1) |

For $\log M_b = 10.45$, $r_M = 6.4791$ kpc, canonical:

| $r$ (kpc) | $x = r/r_M$ | demand $X$ [$M_b$] | $X/5.41$ | where the radius comes from |
|---|---|---|---|---|
| 30 | 4.6303 | 4.157 | 0.768 | COL1 Part 0 |
| **38.15** | **5.8888** | **5.410** | **1.000** | **the whole cosmic budget is spent here** |
| 139 | 21.4537 | 20.96 | 3.874 | FLD1 $R_{\rm fold}(z = 0)$ |
| **272** | **41.9814** | **41.48** | **7.668** | **FLD1 KiDS inner window edge** |
| **637** | **98.3168** | **97.82** | **18.081** | **FLD1 KiDS middle window edge** |

The shortfall is a mass ratio, so it is mass-independent and footing-independent: across the four KiDS masses $X_s/5.41 = 10.99/11.01/10.99/11.01$ canonical, and 12.09 to 12.11 on alt, because the windows scale with $r_M$. Against the delivered charge the gap is a factor $X_s/1.06 = 56.13$; the budget itself is only 5.10 times what arrives.

And the cap is not the record's inference from a fitted share — **KiDS rejects the cosmic share directly**. ONE1's brackets reject $X = 5.41$ at $+4.85/+4.93\sigma$ and PD1 A2's basin share $X = 10.82$ at $+4.34/+4.53$, while accepting 53.64 and 59.5. The survey requires, on its own, a share an order of magnitude above everything that exists.

## 4.4 GAS1, the missing-baryon flank (closed today)

The one reconciliation left inside the record was that $g_{\rm bar}$ at KiDS radii omits hot gas, so the law is being evaluated at the wrong baryons. GAS1 grades that reading against every gas constraint on disk, and returns **MISSING BARYONS EXCLUDED [AMOUNT AND AMPLITUDE]** on both footings.

The gas the reading needs is a fixed surface density,

$$\Sigma_{\rm gas} = \frac{a_0}{\pi G x_b^2}, \qquad x_b^2 = R_b(1 + R_b) = 5.41 \times 6.41 = 34.6781,$$

that is $1.287355 \times 10^{-2}\,\mathrm{kg\,m^{-2}} = 6.1627\, M_\odot\,\mathrm{pc}^{-2}$ canonical and $7.4477\, M_\odot\,\mathrm{pc}^{-2}$ alt: constant in radius, out to 2.6 Mpc, and independent of the galaxy, because it is the column whose enclosed mass makes the law's share exactly $5.41\, M_b$ at that radius. Against the record's own host-free bins it demands $1.33 \times 10^{14} M_\odot$ at the outermost bin, $1913\, M_b$ — every L* galaxy would need the gas halo of a cluster.

Five gates were frozen in advance, each firing only if it holds in every outer bin:

| Gate | What it asks | canonical | alt | Fires |
|---|---|---|---|---|
| G1 | gas needed / **all** baryons in the sphere | 284, 327, 433, 574 | 344, 396, 523, 695 | yes |
| G2 | gas needed / everything measured there | 21.0, 5.43, 2.57, 1.96 | 25.4, 6.56, 3.10, 2.37 | yes |
| G3a | gas needed / **B21's own** hot gas | 3750, 524, 497, 328 | 4532, 633, 601, 397 | yes |
| G3b | gas needed / **GO1's measured** group gas | $+8.0$ to $+13.3\sigma$ | $+10.0$ to $+16.3\sigma$ | yes |
| G4 | outer amplitude *with that gas in* $g_{\rm bar}$ | $0.0392 \pm 0.0069$ | $0.0324 \pm 0.0057$ | yes |
| G5 | required gas density / $\rho_b$ (lower bound) | 1.38 | 1.67 | no |

G4 is the gate that decides it, because it is the only one that closes the loop. The measured outer amplitude of the host-free set is $1.4002 \pm 0.2211/1.2740 \pm 0.2012$ of the law. Put the amount-fixing gas into $g_{\rm bar}$ and the same data sit at $0.039 \pm 0.007/0.032 \pm 0.006$ — 36/41 times below the law, and more than $100\sigma$ below the record's

frozen 0.891 gate line. The reading does not merely need implausible gas; it cannot repair the amount and keep the fit. It is the same gas counted twice.

What survives of the baryon reading is narrow and is stated here rather than dropped: at *cluster* scale the required column is within a factor of about one of what X-COP measures (ratio 0.37 to 1.04 canonical, 0.45 to 1.26 alt), so a baryon reading of the amount stays open for clusters and groups. The *galaxy* leg is closed.

# 5 What is closed, lane by lane

Every lane below froze its criteria in a commit before any graded number was computed, carries known answers that must pass and mutation twins that must fail. "Closed" means the class carries a verdict fixed by frozen criteria; it does not mean a gate passed. Where a frozen reading and a post hoc reading differ, the paper gives the frozen one first.

| Lane | Question | Verdict | Folder |
|---|---|---|---|
| SWT1 | can any action-built switch keep the component out of galaxies? | KEY-DEPENDENT; wall-population class NOT A NO-GO AS FROZEN | fable52_push |
| EMV1 | does the effective-medium allowance survive a nonlinear solve? | T2 HOLDS / T2 HOLDS | fable52_push |
| GAL1 | how much of space may law-on regions fill? | ALLOWANCE CONFIRMED / ALLOWANCE CONFIRMED ($f \leq 0.0197$) | fable52_push |
| KO1 | are KiDS's outer points the lens's own field? | LAW LEVEL HOLDS / LAW LEVEL HOLDS | fable52_push |
| KO1R | the same, regraded blind by another session | reproduces the lane to $2 \times 10^{-15}$ | fable52_push |
| SNP1 | can a shell be cleared and carried dynamically? | FAILS / FAILS | fable52_push |
| HE1 | can the share be held outside a wall, statically? | FAILS / FAILS | fable52_push |
| KO2 | lensing by galaxy type, with neighbours modelled | NO VERDICT for late and early types | closure_2026_11_09 |
| RB1 | groups, clusters and the Bullet | NOT FALSIFIED (ONE-SIDED); never a pass; at the ceiling | closure_2026_11_09 |
| RB2 | hosted dwarfs | B FAILS ON THE HOSTED ROWS | closure_2026_11_09 |
| ONE1 | KiDS host-free lenses, counted once | PASSES / PASSES (does not separate from the law) | fable52_push |
| CO1 | CMB, CMB lensing and growth, computed | ROWS HOLD / ROWS HOLD (inherited halo list) | closure_2026_11_09 |
| CO2 | the amplitude, pinned by large-angle polarisation | AMPLITUDE EXCLUDED BY RAW KiDS AT 3 SIGMA | closure_2026_11_09 |
| KL1 | what carries the amplitude gap | as frozen NO VERDICT; amended AMPLITUDE CONSISTENT WITH RAW LENSING | closure_2026_11_10 |
| KL1R | KL1 Part B, blind regrade | same two readings; blindness not clean | closure_2026_11_10 |
| BX1 | the reference run's realisation noise | as frozen NO VERDICT; interim AMPLITUDE CONSISTENT on one pair | closure_2026_11_10 |
| DG1 | do the high dwarf rows close by history? | HISTORY DOES NOT CLOSE THEM | closure_2026_11_10 |

| Lane | Question | Verdict | Folder |
|---|---|---|---|
| DSB1 | the dwarf classes, full budget frozen blind | STOPPED AT KA-c, no label; amended NO POWER | closure_2026_11_10 |
| SHP1 | the cold component's shape, from raw data | THE LAW'S SHAPE ON ROTATION CURVES, LENSING ABOVE; NOT ONE SHAPE (R3b) | fable52_push |
| DD1 | does one rule cover every class on disk? | NO RULE AMONG THE CANDIDATES | fable52_push |
| HIS1 | the formation reading on DD1's table | HOLDS EXCEPT H3 (fails), H5 (no power) | fable52_push |
| HIS1R | HIS1 regraded blind, separate code | HOLDS EXCEPT | fable52_push |
| FRM1 | the formation reading *as a theory* | HOLDS EXCEPT / HOLDS EXCEPT | fable52_push |
| MK1 | one quadratic dust constant for everything | NO VALUE / NO VALUE | fable52_push |
| MK2 | an exponential dust term | NO VALUE / NO VALUE | fable52_push |
| TK1 | dust stiffness keyed on the expansion | BREAKS AT TURNAROUND | closure_2026_11_10 |
| AKD1 | dust stiffness keyed on the acceleration | BREAKS AT A1 | closure_2026_11_10 |
| AKS1 | a keyed energy sink for the joint | CLOSED, ONE BRANCH ON AN ESTIMATE | toe_2026_11_10 |
| LFS1 | can the leafwise flux supply the halo? | ILL-POSED; a static halo cannot grow | closure_2026_11_10 |
| REL1 | the law on the clock's relative acceleration | NOT WELL-DEFINED | fable52_push |
| COL1 | does a falling clock reach the standing state? | BREAKS DOWN / BREAKS DOWN | fable52_push |
| FLD1 | does the folded region reach the law? | DOES NOT / DOES NOT | fable52_push |
| VSH1 | does a folding flow settle behind a shock? | as frozen NO VERDICT; SETTLES BY SHOCK under post hoc A1 | closure_2026_11_10 |
| EDG1 | can real charge be held in the edge layer? | FAILS AT E3 / FAILS AT E3 | closure_2026_11_10 |
| SET1 | irreversible settling toward the law's halo | STRUCTURE FAILS AT A1 | closure_2026_11_10 |
| CLS1 | is there a cluster state in action (A)? | STATE EXISTS-PARTIAL; NO STATE at the dust constant | closure_2026_11_10 |
| CLS2 | what is the cluster's mass, then? | NO CANDIDATE / NO CANDIDATE | closure_2026_11_10 |
| PAR1 | which parent action could carry the component? | NO VERDICT; no single-valued carrier exists (a posteriori) | closure_2026_11_10 |
| WV1 | wave dust with a split charge | STRUCTURE FAILS | closure_2026_11_10 |
| SMF1 | the simplest smooth-frame parent | STRUCTURE FAILS / STRUCTURE FAILS | closure_2026_11_10 |
| M5 | local, momentum-conserving, normally self-gravitating media | NO-GO, a theorem | rebuild_2026_10_22 |

| Lane | Question | Verdict | Folder |
|---|---|---|---|
| NBC1 | $a_0$ as a contact opacity with neutral gas | KILLED (K2 PROVISIONAL) | fable52_push |
| KDE1 | the clock made into the dark-energy field | ALREADY IN THE RECORD AS AC2s | fable52_push |
| SCS1, SCS2 | is the standing state stable with gravity coupled? | STABLE / STABLE (point masses, isolated discs away from the centre) | fable52_push |
| SCS3 | a disc in a uniform external field | NO VERDICT, with STABLE excluded | fable52_push |
| DCN1 | a disc's own centre | NO VERDICT as frozen; ARTEFACT under six post hoc amendments | fable52_push |
| ELA1 | the medium as an elastic solid | ELASTIC SOLID FAILS AT S0 (+S2, B1) | hy4_preview_push |
| VCC1 | can the clock's charge be sourced instead? | EXCLUDED / EXCLUDED | closure_2026_11_11 |
| GO1 | the baryon census in group outskirts | NOT EVALUABLE [$r_{\rm ta}$ NOT REACHED]; nothing decided | closure_2026_11_10 |
| XCV1 | X-COP without hydrostatic equilibrium | NO VERDICT as frozen; AT THE CEILING under post hoc A1 | closure_2026_11_10 |
| NH2 | NGC 1052's mass against what tides need | NO VERDICT as frozen; DF4 TIDES CANNOT REACH, DF2 NOT DECIDED | closure_2026_11_10 |
| T4 | can polarization charge be made during infall? | exactly 0, any momentum equation | AKS1 part T, SMF1 S0b |
| AMT1 | what is the amount budget? | 11.00× the budget demanded; 56× what arrives | closure_2026_11_10 |
| GAS1 | is the shortfall missing baryons? | MISSING BARYONS EXCLUDED [AMOUNT AND AMPLITUDE] | hy4_preview_push |

Read as one line: **every story about what the cold component is has a failing verdict, and the one number it needs cannot be supplied.** Not "unproven" — excluded, on frozen criteria, one lane at a time.

## 6 The formation reading, and its 4 of 5

What is left is not a substance. DD1 asked the data to name the rule, with no candidate imposed, over 3419 points, 626 systems and 16 classes, scored held-out and leave-one-class-out against a class-constant reference of 0.284 dex. No rule on the record's candidate list holds: not one candidate survives both half-splits in any of 51 seeds, every candidate that covers all 16 classes fails at least 11 of them, and six classes fail every candidate. The verdict is **NO RULE AMONG THE CANDIDATES**.

The reading that does score is not a force law. It is a *history*: the non-baryonic mass is real and conserved, its amount was set when the system formed, and since then a system can only lose it (stripping), which moves it below the law but never below $X = 0$; or lose baryons, which moves it above the law. HIS1 froze five statements and read them on DD1's frozen table. Four hold; the label is **HOLDS EXCEPT H3 (fails), H5 (no power)**. The same run, regraded blind with code written from scratch, gives **HOLDS EXCEPT**.

The **force reading** — the plain law, every system on it now — scores **1 of 5** on the same tests.

Made into a theory and scored pre-registered, the formation reading predicted 14 of 16 class offsets inside their bands before the table was read, including the two the lead named in sign and in size: the stripped families sit low (tidal dwarfs −0.61 to −0.87; DF2/DF4 −0.74 to −0.94) and the hosted classes sit high (MW ultra-faints +0.51

to +0.60 against a predicted +0.30 to +0.70; Coma UDGs +0.31 to +0.37). Its label is **HOLDS EXCEPT / HOLDS EXCEPT**.

It fails where it must, and the failure is structural. Clusters sit **+0.496** (X-COP) and **+0.553** (CLASH) dex above the law, against ceilings of +0.320 and +0.227: 0.18 and 0.33 dex above anything the theory's channels can reach, with 77% and 98% of their points carrying more non-baryonic mass per baryon than the channel's absolute ceiling $a_0/g_{\rm bar}$. Whatever puts the extra mass in a cluster is a *source*, which is exactly what the force reading has and the formation reading does not.

# 7 The decisive measurement, with its pre-registered number

Everything above reduces to one number: the share $X_s = 59.5\,M_b$ that KiDS host-free lenses require. It is 11.00 times the cosmic budget and 56 times what the record's own dynamics deliver, and KiDS rejects the cosmic share at $+4.85/+4.93\sigma$ on brackets frozen before the fit. So the question is not which mechanism supplies the difference. It is whether the difference is a requirement or a census error.

GAS1 turns that into a measurement. The gas that would bring the law's demand inside the cosmic budget is a fixed column,

$$\Sigma_{\rm gas} = \frac{a_0}{\pi G x_b^2} = \frac{9.3603\times10^{-11}}{\pi\times6.674\times10^{-11}\times34.6781} = 1.287355\times10^{-2}\,{\rm kg\,m^{-2}} = 6.1627\,M_\odot\,{\rm pc^{-2}},$$

canonical, and $7.4477\,M_\odot\,{\rm pc^{-2}}$ alt; constant in radius out to 2.6 Mpc; independent of the galaxy.

**The pre-registered acceptance line is 0.891** $= 10^{-0.05}$ — the record's KiDS outer-amplitude gate (KO1's LINE), written into GAS1's criteria before any script or number. The reading needs a gas column of $6.16/7.45\,M_\odot\,{\rm pc^{-2}}$ at 0.1 to 2.6 Mpc to make the amount balance, and that same gas in $g_{\rm bar}$ puts the outer amplitude at $0.039\pm0.006/0.032\pm0.006$: 36/41 times below the line.

So the decisive measurement is named, and it has a number:

> Measure the hot-gas mass at 0.1–2.6 Mpc around an *isolated* L* galaxy. The framework's reconciliation needs $\Sigma_{\rm gas} = 6.16/7.45\,M_\odot\,{\rm pc^{-2}}$, i.e. $1.33\times10^{14}M_\odot$ inside 2.6 Mpc. If the column is **not** there, form B's halo is not the cold charge.

No such measurement exists in this repo. X-COP's gas profiles reach those radii at *cluster* scale and give a column within a factor of about one of the requirement; at galaxy scale nothing is on disk. Go and take it.

# 8 Verdict: what survives, and what to do

**What survives.**

1. **The $a_0$ relation as a measurement.** 0.057/0.021 dex over 2700 rotation-curve points in 149 galaxies, and a frozen-gate pass on KiDS host-free lenses at $1.4002\pm0.2211/1.2740\pm0.2012$ of the law's level. Nothing in this note touches it, and no lane on the record has broken it.
2. **The halo is real, separately conserved, non-negative mass.** 0 of 626 systems read below $X = 0$ beyond three of their own errors.
3. **The standing-clock state is a genuine solution of the adopted action**, it holds the law's shape in any geometry by an exact force-balance identity, and it is linearly stable with gravity coupled around point masses and around isolated discs away from their centres.
4. **The formation reading scores 4 of 5** against the force reading's 1 of 5, and holds up as a scored theory: 14 of 16 classes inside pre-registered bands.

**What is closed.** Every candidate for what the cold component *is* — cold scalar dust, a dark-energy clock, a warm component, a superfluid, a dipolar medium, a tethered medium, two phases, a medium in momentum contact with baryons, a parent action built on a single-valued carrier, a smooth frame, a spatial switch, a relative-acceleration law, keyed dissipation, the leafwise flux, an elastic solid, a sourced charge, a contact opacity. Not "unfalsified": each has a failing verdict on frozen criteria (Section 5).

**What to do.**

1. **Stop deriving the law from a substance.** That programme has an exhaustive negative result now, and running more of it is theatre.
2. **Derive the *amount* from formation instead.** Fix the epoch and the conserved quantity. The pre-registered prediction to test: stripped systems (tidal dwarfs, DF2/DF4) sit LOW, hosted systems sit HIGH, and within a class nothing tracks a measured sign of history. That last one is the reading's signature, not a hole in it.
3. **Take the one measurement that decides the galaxy leg** — the hot-gas column at 0.1–2.6 Mpc around an isolated L* galaxy, against the pre-registered line 0.891. A census that falls short falsifies form B on the galaxy leg outright. At cluster and group scale the same column is *not* excluded, and that leg stays open.
4. **Keep what is cheap to kill in reach**: Gaia DR4 wide binaries (2026-12-02), scored as pre-registered; a group census from $R_{500}$ to the turnaround radius with a lensing mass to that radius; a cluster mass beyond $R_{500}$ that does not assume hydrostatic equilibrium; the next lensing release, where the amplitude gap has to stay inside two widths.

## 9 Our own estimates that were wrong

Recorded before the numbers that tested them, and kept here rather than dropped. (i) The CRUX bet: GAS1 would find the gas census closes at least half of the 11.0×. **Lost**, by 36 to 41 times on the amplitude gate. What survives of it is narrower: the column the *cluster* rows need is within a factor of about one of what X-COP measures, so the baryon reading is open at cluster and group scale and closed at galaxy scale. (ii) The proposal that the high dwarf rows are the ones that lost their gas: the class medians matched and the per-galaxy gates failed (DG1). (iii) The acceleration key as a stiffness with a healthy gradient sign: AKD1 found the sign the other way. (iv) The expectation that the standing state's instability survives coupling to gravity: it does not; it belongs to a planar background (SCS1). (v) CO2's hand estimate of the optical depth (0.055 to 0.060): it came out just below, at 0.0532.

## Method

Sources, all in this repository and all checked by `PAPER73_audit.py`: `fable52_push/` (HIS1, HIS1R, SHP1, DD1, ONE1, SWT1, EMV1, GAL1, KO1, KO1R, SNP1, HE1, MK1, MK2, REL1, COL1, FLD1, SCS1, SCS2, SCS3, DCN1, NBC1, KDE1, ELA1), `closure_2026_11_09/` (KO2, RB1, RB2, CO1, CO2, PD1, PD2), `closure_2026_11_10/` (LFS1, SMF1, PAR1, WV1, TK1, AKD1, KL1, KL1R, BX1, DG1, DSB1, GO1, XCV1, NH2, SET1, EDG1, VSH1, SRH1, AMT1, R541), `closure_2026_11_11/` (VCC1, DW1), `toe_2026_11_10/` (AKS1 part T), `rebuild_2026_10_22/` (M5, PH1, IJ1, HF1, TM1, X38), and the assembly notes `closure_2026_11_10/THEORY_ASSEMBLED_2026-11-10.md`, `hy4_preview_push/CRUX.md`, `hy4_preview_push/CLOSURE.md` and `hy4_preview_push/EXITS.md`. None of these lanes is registered in the closure record and none relabels an earlier lane; where a lane's own RESULT.md and this note differ, the lane is right.

Four naming rules hold throughout. (a) Cold dark energy is the clock's dust: a gravitating field component with $w \approx 0$, never "cold dark matter" and never a particle. (b) The footings are never pooled. (c) $\kappa = \frac{1}{2}$ is stated as fitted everywhere it is used, and the theory is stated as not closed. (d) Post hoc items are called post hoc in the lane that made them; where a frozen reading and a post hoc reading differ, the frozen one is given first.

**T4 has no lane folder of its own.** It is a theorem, and its two records are `toe_2026_11_10/AKS1_acceleration_keyed_sink/FROZEN_CRITERIA_T.md` ("T4 GAUSS (exact; SMF1 S0b reproduced)") and `closure_2026_11_10/SMF1_smooth_frame_parent/RESULT.md` ("S0b (exact, Gauss)"). AKS1's part T is post hoc; its criteria were committed alone before its script.

**What is from memory.** No physical number in this note is from memory. Every quoted value is read from a committed file in this repository and re-checked, literal by literal, by `PAPER73_audit.py`; the script also recomputes the two ratios that are the paper's own arithmetic. What *is* from memory is context, not data: the record's conventions (footings never pooled, $\kappa = \frac{1}{2}$ fitted, the theory not closed, what each lane label means), and this session's reading of which lanes carry which verdict. The lane assignments in Section 5 follow the assembly notes named above; where they and a lane disagree, the lane is right.

**Limits.** The KiDS numbers rest on post hoc amendments in several lanes, as each of those lanes says. The measurement of Section 7 does not exist on disk; this note states what it would have to find, not what it finds.

The formation reading's epoch is not fixed, and the cluster failure in Section 6 is unresolved by everything on the record.